The reader and the rim: the detection rule varied, and a birth phase tied to the approach phases

Supporting computations for arXiv:2608.18886 — deposit note, version 6

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Setting

Versions 3–5 characterized the floor-attaining census $\rho\left(\xi_A, \xi_B, \tau, s\right)$ with the detection rule fixed at $A = \operatorname{sgn}\cos 2\left(\lambda - a\right)$ and the pair's initial phase λ_0 uniform and independent. This version varies both.

The reader

The λ-histograms of every $(\tau, s, \text{setting}, \text{offset})$ were stored (32×32 lattice; the step reproduces the deposited banks to 2×10^{-15}) so that any detection profile $D\left(\theta\right)$, $|D| \le 1$, is a re-contraction. Thirty-two profiles: the step; the smooth $\cos 2\theta$; blunted steps of edge width 10.6–64°; domed and tilted caps; notched caps (and the twin carrying the complementary notches); front/back walls of different width (breaking the port rule) and the twin mirrored; kinked walls; walls whose position follows the approach phase; staircase walls of 2–8 risers and 12–36° width.

Results (MD/floor at $\varepsilon = 0.01$, full census): the step 1.0000; a smooth edge of width 10.6 / 12.1 / 14.2 / 17.1° costs 1.048 / 1.169 / 1.457 / 2.351 and 21.5° loses the singlet; every cap feature (dome, tilt, notch) loses the singlet at any census bandwidth (miss linear in the feature's size); complementary notches on the twins are excluded by an identity ($|E| \le \left(1 - E/2\right)^2$; computed 0.2345 vs 0.2344); a front/back-asymmetric wall is priced by its wider wall and polarizes the singles; a wall following the approach phase posts a λ-marginal price below the floor that is not a price (the joint hidden variable's MD is 6–7× the floor); a staircase wall of 12° prices 1.000–1.005 whatever the riser count while the smooth wall of the same width prices 1.17 — at equal lost area, sharp-edged loss is far cheaper than graded loss, and about 6° of sharp-edged loss per edge is free (a three-level reader undecided within ±3° of the boundary attains the floor exactly; ±4.5° costs 16%, ±6° 94%, ±9° loses the singlet). No reader lowers the band-limited prices: every shaped reader raises them ($K = 6$: 2.2–2.6 against the step's 1.70). The reader is a mesa: flat cap, no slope, kerf ≤ 6°, twins alike, blind to the approach phase.

The floor at $\varepsilon = 0$

The price LP with a zero-width correlator band (322 equalities) is feasible at MD/floor 1.0000.

The axis

On the terrain cut to station indices ≤ 3 the sum-lock pixel (2,2) stays on top with its axis within 3° of the full terrain's; cut to ≤ 2, (4,4) takes over (period 45°), the Alice marginal reverses phase, and a relative-phase pixel rises to $z = +3.7$. With the October perturbation run, the axis belongs to the (2,2,+) sum beat kept on top by the order-3 difference beats.

The rim

Every initial particle's final bin was stored (same trajectories; histograms reproduced bit for bit) so that the birth may be a joint density $w\left(\xi, \tau, s, \lambda_0\right) = \sum_l w_l\left(\xi, \tau, s\right) f_l\left(\lambda_0\right)$ with L harmonics in λ_0 and the usual band limit K in ξ,

positivity of the joint density, mass, CHSH exact, band $\varepsilon = 0.01$, MD on the $\lambda\,(\,\tau\,)$ marginal. Prices (MD/floor), each bracketed by the LP value and a clipped re-measured census agreeing to 10^{-4}: uniform λ_0: 1.700 / 1.429 / 1.258 at $K = 6$ / 8 / 10 (the deposited values). Rim depending on the dial only: identical to four digits. Rim depending on the approach phases, one harmonic: **1.373** at $K = 6$, **1.164** at $K = 8$; two harmonics: **1.230** at $K = 6$. Rim restricted to degree 3 over the approach phases: 1.647 / 1.412 — the gain is carried by the rim's fine structure.

The fitted rim is a half-wave (relative modulation 0.95 at half the atoms; depth 0.86); its lean's Fourier skeleton lies entirely on pixels with $k_A + k_B$ odd (the census's on even: an exact consequence of the model's +90° twin symmetry) with $(\,-1, 0\,)$ leading at 0.54–0.62 in all three solutions (null 0.06; same phase within 9°): the preferred birth phase is $\lambda_0^* \approx 77\text{–}81° - \xi_A$ in the pinned frame; with two harmonics the second lives on the even pixels and is keyed to the other station ($(\,0, \pm 2\,)$ leading), and the shallow cells carry a two-lobed rim where the deep cells carry one lobe. This enlarges the model class of the paper (λ_0 uniform and independent); the floor is unchanged, the census's bandwidth demand is what moves.

Negatives

$K = 20$ on the 4096 census: stopped after 208 h in the interior-point solve (the curve ends at $K = 18$, 1.012). Seven joint-birth LPs at the census's degree with $K \geq 8$ or $L = 2$ failed ("numerical difficulties" after 41 h) or were stopped at 47–100 h; the lighter formulation of `census_fuzz_v3` solved the $K = 6$ / $L = 2$ and $K = 8$ / $L = 1$ cases in 21 and 9 h.

Scope

One terrain; lattice sampling (the appropriate sampling for smooth-structure prices; the rim's fine structure awaits random offsets); pinned frame; $\varepsilon = 0.01$ unless stated. LP prices are exact optima re-verified from the reconstructed census; the joint-birth prices are bracketed as stated.

Files

Scripts `pipeline_bulge_v1.py`, `census_profile_v1.py`–`v4.py`, `run_bulge2_v1.py`, `pipeline_fuzz_v1.py`, `census_fuzz_v1.py`–`v3.py`, `rim_skeleton_v1.py` and launchers; transcripts `bulge_*.txt`, `bulge2_*.txt`, `fuzz_*.txt`, `fuzz2_*.txt`, `fuzz3_*.txt`, `rim_skeleton_*.txt`, the pipeline consoles, and `bl_n64_k20.txt` as it stands; the histogram and particle stores (`hist_*.npz`, `part_*.npz`); the `census_profile_*` and `census_fuzz_*` JSON outputs and saved solutions; the dated notes and the two pre-registrations under `notes/`.